Pa Tools hardware update

Date: 2026-10-24 **Branch:** `claude/hardware-m1` (pushed, head 1e009b4) **Base:** `pa-tools` d1c5e6f

Everything below was compiled and tested in a Linux cloud container with no audio or MIDI hardware. Nothing has been run against a real interface, keyboard or AAF file. Test results were re-run by the coordinator, not only reported by agents.

What landed (5 commits)

Gap	Delivered	Not done
G-01 audio interface	Latency math and measurement from stream timestamps; Player/Recorder latency accessors; opt-in record-latency compensation applied when a take becomes a clip (default off); Setup checkbox; command `setup.audio_latency`	True duplex on one clock (cpal does not offer it); channel routing / I/O mapping; record-to-disk (G-06); ASIO in official builds; any real-device test
G-02 MIDI devices	Port listing, live input (non-blocking bounded queue, drop counting) and output via the MIT `midir` crate, no unsafe; strict message parser; tick conversion; pure MIDI take recorder producing `soundcraft-midi` sequences; command `setup.midi_ports`	Recorder is not connected to the transport or armed tracks; no live monitoring/thru; no clock/MTC; no real-port test
G-03 session interchange	New crate `soundcraft-interchange` : hostile-input-safe Compound File reader and AAF identification, 12 tests	AAF composition parsing, import into sessions, export, OMF. No real AAF sample was available, so AAF structure details are unverified

Verification (coordinator run)

Check	Result
cargo fmt --check	**clean**
clippy -D warnings (playback, engine, ui-egui, interchange)	**clean**
Tests: playback 73, engine 261+, interchange 12	**pass**
cargo xtask layers	**ok**
UI crate tests	Agent reported 121 lib tests pass; `theme_docking` cannot run here (no GPU); one recovery test was flaky once. Not independently re-run
cargo xtask assets	**fails on two existing docs images missing from ATTRIBUTION.md (not caused by this work)**
Real hardware	**none available; all device paths unverified**

Corrections to earlier planning

- The earlier handover proposed hand-written CoreMIDI/WinMM/ALSA unsafe bindings. AGENTS.md forbids unsafe outside three plugin-host crates, so `midir` was used instead.
- Earlier hour estimates and cost figures were not measured and should be disregarded.

Next work (needs real hardware or samples)

1. Test latency and MIDI ports on a Mac with a real interface and keyboard.
2. Connect MidiTake to armed MIDI tracks and the transport.
3. Obtain real AAF files, then parse compositions and map to sessions.
4. Streaming record-to-disk (G-06).
5. Channel routing in I/O Setup.

Rule check: nothing was written to Gil's drives; all work is in the repository.